



Department of Information Technology

Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch : IV Semester-B. Tech Information Technology
Course Code & Title : IT3401 & Web Essentials (Integrated Course)
Name of the Faculty member (s) : S. Sakkaravarthi, AP/IT

Innovative Practice Description

- **Unit / Topic** : Unit 5 / Cookies
- **Course Outcome** : CO5
- **Topic Learning Outcome:** PrO5
- **Activity Chosen** : Learning by Teaching
- **Justification:**
 - Learning by Teaching is an innovative practice adopted to teach the topic Cookies in Java Servlets, wherein students assume the role of instructors to explain session management concepts to their peers.
 - This method promotes active learning, enhances self-confidence, deepens the understanding of servlet-based cookie handling, and fosters collaborative knowledge building in the context of web application development.
- **Time Allotted for the Activity:** 35 Minutes
- **Details of the Implementation:**
 - To implement the Learning by Teaching method for the topic Cookies in Java Servlets, the instructor invited students to volunteer for teaching roles. Students Alagusurya S and Kirran S T, showed interest.
 - Based on their willingness, the two students were assigned specific subtopics related to Cookies, such as the purpose of cookies in session management, how to create and retrieve cookies in servlets.
 - To ensure adequate preparation, the topics were shared with them 3 days before the session. This gave them sufficient time to organize their content and teaching strategies.
 - During the session, the students delivered their assigned subtopics to the class with clarity and enthusiasm. First, the student Alagusurya S explained the purpose of cookies in session management and real world-application such as remembering login credentials or user preferences across sessions. Followed by the student Kirran ST explained how to create and retrieve cookies with suitable client-server interaction diagram.



- The session encouraged interaction by allowing students to ask questions during and after the presentations. This peer-driven discussion helped reinforce core concepts of cookie-based session management in Java Servlets and promoted a deeper understanding of the topic.
- Finally, the instructor summarized the content.

○ **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO10	PSO3
CO5	2	2	3	2	2	3

(1 – Low 2 – Moderate 3 – High)

○ **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO5	PO10	PSO3
CO5	2	2	3	2	2	3
Justification for correlation	Students will develop a clear understanding of cookie-based session tracking in web applications and apply engineering concepts to real-world web development scenarios.	Students will be able to demonstrate their ability to explain cookie implementation, showcasing their skill in analyzing how web servers manage user sessions.	Students will be able to implement session management functionality in web applications using java servlet cookies	Students gained proficiency in using software tools and technologies relevant to web application development	Students can improve their ability to communicate technical concepts effectively.	Students will be able to implement session tracking using Java Servlet cookies to meet real-time business requirement in modern web applications.



○ **Images / Screenshot of the practice:**



Fig 1: Learning by Teaching Activity by Alagusurya S and Kirran ST

○ **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- By taking on the teaching role, students felt more confident and gained a deeper insight into session management using cookies in web application development.
- All students listened attentively to the session and felt that the topic was clearly understood through peer teaching.
- The activity enhanced student's understanding of cookie-based session management in java servlet which is essential for creating interactive and user-focused web applications.

❖ **Benefit of the practice:**

- Students were more attentive and involved, both as presenters and audience members, resulting in a more lively and engaging learning environment.
- This activity helped students enhance their communication skills and self-confidence through peer teaching.
- Students were able to connect technical concepts to real-world applications such as login systems and shopping carts, making the topic more meaningful and easier to remember.



❖ ***Challenges faced in implementation:***

- Some Students lacked foundational understanding of HTTP Statelessness and servlet lifecycle making it difficult to grasp how cookies work in session management.
- Few Students used incorrect syntax while creating Cookie objects or forget to pass the required parameters like name and value.
- This activity required extra time and effort compared to regular teaching.

At the end of session, the instructor recapped HTTP Statelessness and emphasized the need of cookies to maintain user identity in web applications. Also to address mistakes in cookie creation, the instructor provided correct syntax and explained how to properly pass the name and value.

References:

- <https://effectiviology.com/protege-effect-learn-by-teaching/>
- https://www.ritrjpm.ac.in/images/computerscience/20222023/5_VA_CS3362_Learning%20by%20teaching.pdf

Signature of Faculty Member

HOD